THE UNIVERSITY OF BRITISH COLUMBIA
Physics and Astronomy
Faculty of Science

Department of Physics
RWTH AACHEN UNIVERSITY

University of British Columbia
RWTH Aachen University

Rotational Dynamics of Molecules in Superfluid Helium Nanodroplets Driven by an Optical Centrifuge

Master Thesis

Sören Eike Mahr

Matriculation number: 466949

First examiner: Prof. Dr. Markus Ternes
Second examiner: Prof. Dr. Valery Milner

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Abstract

[TODO: English abstract.]

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1 Introduction

1.1 Motivation

[TODO: Why study molecular rotation in superfluid helium nanodroplets.]

1.2 State of the Art

[TODO: Existing work on laser-controlled rotation and doped droplets [1].]

1.3 Objectives and Outline

[TODO: Research question and chapter-by-chapter outline of the thesis.]

2 Theoretical Background

2.1 Superfluid Helium Nanodroplets

[TODO: Superfluidity, droplet temperature and size, doping via pickup.]

2.2 Molecular Rotation

The rotational energy of a linear rigid rotor is

$$E(J) = B\,J(J+1), \tag{2.1}$$

with rotational quantum number J and rotational constant B. [TODO: Angular momentum, selection rules, gas phase vs. in-droplet rotation.]

2.3 Molecules in Helium Nanodroplets

[TODO: Reduced effective rotational constant, coupling to the superfluid.]

2.4 The Optical Centrifuge

[TODO: Field construction, accelerating polarisation, angular trapping, adiabatic climbing of the rotational ladder.]

3 Methods

3.1 Experimental Setup

[TODO: Overview of the apparatus.]

3.2 Optical Centrifuge Generation

[TODO: Pulse-shaping and field-generation scheme.]

3.3 Detection and Data Acquisition

[TODO: Detection method and data acquisition.]

3.4 Data Analysis

[TODO: Analysis procedure and processing pipeline.]

4 Results and Discussion

4.1 Result Set A

[TODO: First set of results with figures and analysis.]

4.2 Result Set B

[TODO: Second set of results.]

4.3 Discussion

[TODO: Interpretation, comparison with theory and prior work.]

5 Conclusion and Outlook

5.1 Summary

[TODO: Summary of the main findings.]

5.2 Outlook

[TODO: Open questions and possible future experiments.]

A Additional Material

[TODO: Appendix content.]

Bibliography

[1] J. Doe and J. Smith. “An Example Reference”. In: *Journal of Example Physics* 1.1 (2024), pp. 1–10. DOI: `10.0000/example`.

Acknowledgements

[TODO: Acknowledgements.]

Declaration of Authorship

I hereby declare that this thesis is my own work and that I have used no sources or aids other than those indicated. All passages taken verbatim or in substance from published or unpublished sources are marked as such. This thesis has not been submitted, in the same or a similar form, to any other examination authority.

Place, Date	Signature